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A Search for Anomalous Spectral Energy Distributions in CEERS DR1: An Open-Source JWST Anomaly Detection Pipeline

Armeen Shasti-Nazem[1]

[1] *Department of Astronomy, University of Washington, Seattle, WA 98195, USA*

JWST NIRCam surveys now provide photometric catalogs large enough to search systematically for galaxies whose SEDs are not well described by standard stellar-population templates. We present an open-source pipeline that fits photo-z's and best-fit template SEDs with EAZY (Brammer et al. 2008), scores per-band residuals with an Isolation Forest + UMAP/DBSCAN ensemble, and cross-matches flagged sources against known contaminants, applied to the CEERS Data Release 1.0 catalog (Cox et al. 2025).

Of 87,345 CEERS DR1.0 sources ingested, redshift-window ($0.5 \leq z_{\rm phot} \leq 10$) and coverage cuts retain 68,839 for EAZY fitting; its fit-failure sentinel ($\chi^2_{\rm EAZY} < 0$) excludes 2,900 (4.2%) as non-converged artifacts, leaving a clean sample of 65,939 sources, of which the top 2% (1,319) are flagged as anomalous. Cross-matching against Milliquas v8 (Flesch 2023) within 1.″5 finds only 5 matches (0.38%): known AGN explain a negligible fraction of the flagged population.

Figure 1 bins the flagged rate in $\Delta z = 0.5$ shells; an OLS fit finds no significant redshift trend ($p = 0.32$). Following the conservative Bayesian framework of Wright et al. (2014) and Lingam & Loeb (2019), we adopt a prior $P(H_1) = 10^{-8}$ per source and the flagged sample's unexplained fraction as $P(\text{unexplained} \mid H_0)$; the posterior that an unexplained flagged source reflects non-standard physics is $P(H_1 \mid \text{unexplained}) \approx 1.12 \times 10^{-8}$, almost unchanged from the prior, since being unexplained is the norm here — a reproducible upper limit on non-standard astrophysics at current CEERS depth.

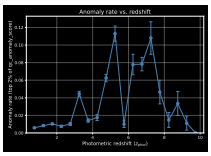


Figure 1. Flagged rate vs. z, clean sample ($N = 65{,}939$), Poisson error bars; rates span ∼0–11%; $p = 0.32$.

Code: https://github.com/Menrae/unsupervised-anomaly-detection-pipeline.

Software: astropy, eazy-py (Brammer et al. 2008), scikit-learn, umap-learn, xarray

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aasha@uw.edu